

Erwin Quanten

Staff Software Engineer – .NET, Azure/AWS

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PROFESSIONAL SUMMARY

Senior .NET Backend Engineer with 16+ years of experience building enterprise platforms across cloud infrastructure, digital consulting, chemicals, manufacturing, logistics, and internal business systems, using **C# 3.0–12**, **.NET Framework 3.5–4.8**, **.NET Core 2.1–3.1**, **.NET 5–8**, **ASP.NET Core 2.1–8**, **Web API**, **EF Core 2.1–8**, **SQL Server 2008 R2–2022**, and **Azure DevOps**. Strong background in migrating legacy monoliths to modular APIs, improving API latency by 22–34%, reducing production defects by 18–27%, and delivering secure backend services with practical architecture, testing, observability, CI/CD discipline, and production support ownership.

SKILLS

- **Backend:** **C# 3.0–12**, **.NET Framework 3.5–4.8**, **.NET Core 2.1–3.1**, **.NET 5–8**, **ASP.NET Core 2.1–8**, **Web API**, Minimal APIs, MVC, WCF, Windows Services
- **Database:** **SQL Server 2008 R2–2022**, PostgreSQL, T-SQL, Stored Procedures, Query Optimization, EF 5–6, **EF Core 2.1–8**, Dapper
- **Cloud & DevOps:** **Azure**, Azure App Service, Azure Functions, Azure Service Bus, Azure Key Vault, Azure DevOps, GitHub Actions, Docker, CI/CD Pipelines
- **Architecture:** REST APIs, Microservices, Modular Monolith, Clean Architecture, CQRS, DDD, Repository Pattern, Dependency Injection, Event-Driven Architecture
- **Testing & Quality:** xUnit, NUnit, Moq, FluentValidation, Integration Testing, Contract Testing, SonarQube, Code Review, Debugging, Production Support
- **Security & Observability:** OAuth2, JWT, RBAC, OWASP, Application Insights, Serilog, Structured Logging, Health Checks, Monitoring, Alerting

EXPERIENCE

Lightedge — Lead Software Engineer

Remote/ Full time

May2022 – Apr 2026

Houston, TX

- Led backend modernization for cloud infrastructure and managed-service platforms using **C# 10–12**, **.NET 6–8**, **ASP.NET Core 6–8**, **EF Core 6–8**, **SQL Server 2019–2022**, and **Azure**, improving API response time by **31%** across customer provisioning workflows.
- Designed secure REST API services for billing, infrastructure inventory, customer onboarding, and internal operations, reducing manual backend support effort by **24%** through stronger validation, workflow automation, and reusable service contracts.
- Migrated several legacy **.NET Framework 4.7–4.8** services into **.NET 6–8** APIs with dependency injection, health checks, background workers, and structured logging, reducing deployment failures by **21%**.
- Resolved recurring timeout issues in high-volume SQL procedures by rewriting joins, adding targeted indexes, removing redundant ORM calls, and lowering average query execution time by **38%**.
- Implemented **Azure Service Bus** messaging between provisioning, notification, billing, and audit services, improving backend reliability when downstream systems were slow or temporarily unavailable.
- Built authentication and authorization layers with **JWT**, role-based access control, claims validation, and secure configuration through **Azure Key Vault**, reducing security review findings by **26%**.
- Created background job processors with **.NET Worker Services**, retry policies, dead-letter handling, and idempotent processing to prevent duplicate customer actions during infrastructure automation failures.
- Improved production troubleshooting by adding **Serilog**, correlation IDs, distributed tracing, and Application Insights dashboards, helping engineers identify root causes faster during service incidents.
- Introduced xUnit integration tests, contract tests, and mocked infrastructure dependencies around critical API flows, reducing regression bugs by **23%** before release.
- Refactored large controller-heavy APIs into cleaner service, repository, validator, and mapping layers, making feature delivery more flexible for customer-specific infrastructure requirements.
- Partnered with DevOps engineers to build **Azure DevOps** and GitHub Actions pipelines with automated builds, test gates, container packaging, and environment-based deployments.
- Mentored backend engineers on **Clean Architecture**, async programming, database optimization, and production-safe release practices while maintaining ownership of critical backend services.

Syscom Digital — Staff Software Engineer

Remote/ Full time

Nov 2017 – Apr 2022

Bucharest, Romania

- Delivered enterprise backend systems for digital consulting clients using **C# 7–10**, **.NET Core 2.1–3.1**, **.NET 5–6**, **ASP.NET Core**, **EF Core 2.1–6**, **SQL Server 2016–2019**, and Azure services.
- Migrated customer-facing APIs from **ASP.NET MVC 5** and **Web API 2** into **ASP.NET Core 3.1–6**, improving maintainability and reducing endpoint latency by **28%**.
- Designed multi-tenant backend modules for user management, reporting, document workflows, and service integrations with flexible configuration for different client environments.
- Solved production memory-leak issues caused by unmanaged dependency lifetimes and excessive object creation by improving DI scopes, caching rules, and async database access.
- Implemented message-based integrations with **Azure Service Bus** and background workers, reducing failed third-party synchronization jobs by **19%** during peak traffic periods.
- Built secure authentication flows with **OAuth2**, **JWT**, refresh tokens, and claims-based authorization for enterprise portals handling role-sensitive business data.
- Improved database performance by tuning stored procedures, replacing inefficient EF queries, and introducing read-optimized projections, cutting report generation time by **33%**.
- Created reusable backend libraries for validation, exception handling, audit logging, API responses, and pagination, increasing delivery consistency across multiple client projects.
- Supported CI/CD modernization through **Azure DevOps**, automated test execution, environment variables, secret handling, and controlled releases across development and production.
- Reviewed backend designs, guided engineers on API contracts and error handling, and helped teams reduce recurring production bugs by **22%** through stronger engineering standards.

Tessenderlo Group — Senior Software Engineer

Remote/ Full time

Aug 2015 – Oct 2017

Brussel Metropolitan Area

- Built backend services for chemicals, manufacturing, supply-chain, and finance operations using **C# 5–7**, **.NET Framework 4.5–4.7**, **ASP.NET MVC 4–5**, **Web API 2**, **WCF**, **EF 6**, and **SQL Server 2012–2016**.
- Modernized internal business applications by replacing tightly coupled Web Forms and service logic with layered **ASP.NET MVC** and **Web API** components.
- Reduced production incident volume by **25%** after fixing null-reference errors, transaction rollback gaps, missing validation rules, and inconsistent exception handling patterns.
- Designed backend modules for order processing, plant reporting, inventory visibility, approval workflows, and finance integrations used by multiple operational departments.
- Optimized SQL Server jobs and reporting queries by improving indexes, removing cursor-heavy logic, and separating read-heavy operations from transactional write flows.
- Integrated internal systems with ERP and third-party services through **WCF**, **SOAP**, scheduled jobs, file-based imports, and carefully controlled database synchronization processes.
- Improved deployment stability by documenting configuration steps, separating environment settings, and creating repeatable release checklists for QA and production environments.
- Added audit trails, role checks, and validation layers around sensitive operational workflows, reducing manual compliance verification effort by **18%**.
- Supported junior developers through code reviews, debugging sessions, and practical guidance on **C#**, **SQL**, service boundaries, and maintainable enterprise application design.

Tessenderlo Group — Software Engineer

Hybrid/ Full time

Sep 2011 – Jul 2015

Brussel Metropolitan Area

- Developed internal enterprise applications using **C# 4–5**, **.NET Framework 4.0–4.5**, **ASP.NET MVC 3–4**, **WCF**, **ADO.NET**, **Entity Framework 5**, and **SQL Server 2008 R2–2012**.
- Implemented backend features for procurement, warehouse operations, production tracking, and employee self-service systems with strong focus on reliability and data accuracy.
- Resolved recurring integration errors between internal applications and ERP exports by improving validation, transaction handling, retry logic, and reconciliation reports.
- Built reusable service layers and repository patterns to reduce duplicated data-access logic and improve maintainability across multiple business applications.
- Improved SQL query performance by rewriting inefficient joins, normalizing reporting tables, and adding indexes that reduced several slow reports by **29%**.

- Supported migration from older ASP.NET Web Forms screens to **ASP.NET MVC** modules while preserving business rules and minimizing disruption for operational users.
- Created scheduled Windows Services for batch processing, notifications, imports, and data cleanup tasks, reducing manual administrative work by **21%**.
- Worked closely with business analysts and plant operations teams to translate practical workflow issues into stable backend features and supportable releases.

Tessenderlo Group — Software Developer

Hybrid/ Full time

Jun 2009 – Aug 2011

Brussel Metropolitan Area

- Developed early-stage business applications using **C# 3.0–4.0**, **.NET Framework 3.5–4.0**, **ASP.NET Web Forms**, **ADO.NET**, **WCF**, and **SQL Server 2008 R2**.
- Fixed production bugs related to incorrect form validation, broken stored procedures, session-state issues, and data-entry edge cases in internal workflow systems.
- Created database-driven modules for internal reporting, user administration, document tracking, and approval routing across finance and operations departments.
- Improved maintainability by extracting repeated code into shared helper classes, service methods, and stored procedure patterns used across several applications.
- Supported senior engineers during upgrades from older .NET applications toward more structured service-based architecture and cleaner database interaction patterns.

EDUCATION

KU Leuven — Master's Degree, Computer Science

Sep 2003 – May 2009